

Note: tab = 8 spaces, '\$'=24H

```
str db 35h, 30h, 33h, 24h
mov ah, 9
lea dx, str
int 21h
mov bx, dx
add bx, 3
mov di, [bx]
mov ah, 2
int 21h
```

```
str db 'Assembly', 10, 10, 'Final', 13, 9, 'EXAM', '$'
mov ah, 2
mov bh, 0
mov dx, 0104h
int 10h
mov ah, 9
mov dx, offset str
int 21h
```

A large rectangular grid consisting of 10 columns and 4 rows of squares, intended for drawing a picture.

```

3) X db 'success',10,13,'is',10,13,'every',10,13,'things'
mov ah,9
mov dx,offset x
int 21h
mov ah,6
mov al,2
mov cx,0
mov dx,0102h
int 10h

```

7. Assuming AL = -1 and BL = -1, the instruction: DIV BL will produce: a) AX = 100H b) AX = -1 c) AX = 0001H d) AX = 00FFH	8. TEST instruction is equivalent to: a) shr b) jnz c) shl d) cmp
9. org 100h mov cx, 3 mov si, 0 li: mov al, x[si] shl al, 2 mov x[si], al inc si loop li hlt x db 2, 4, 6 a) X=08H, 10H, 12H b) X=08H, 04H, 06H c) X=02H, 04H, 06H d) X=01H, 04H, 0CH	10. ORG 100h MOV BX, OFFSET X PUSH [BX+1] MOV AX, [BX] POP CX HLT X DW 4020H, 22D5H, 0AAFFH a) AX= D540H CX=4020H b) AX=4020H CX=D540H c) AX=2040H CX=D530H d) AX=4021H CX=4020H
11. The value of BH after executing the following instructions is ____h STC MOV CL, 2H MOV BH, 14 RCL BH, CL a) 32 b) 19 c) 3A d) 1D	12. To enter single character from the keyboard using int 21h we use subfunction ____h a) 09 b) 0A c) 01 d) 02
13. The value of x after executing the following code is ____h X DW ? MOV AX, 0B2BH PUSH AX INC AX PUSH AX POP X a) 0B2B b) 0B2A c) 0A2C d) 0B2C	14. The value of Y after executing the following code is ____ X DB 'Assembly' Y DB 5 DUP(*) STD MOV SI, OFFSET X + 6 MOV DI, OFFSET Y + 4 MOV CX, 3 REP MOVSB MOV AX, 7H STOSB a) *#mbi b) 7#mbi c) scmbi d) 7#scmbi

21. To convert AL from binary to ascii : ex: 9 to '9' a. AND AL,0FFH b. OR AL,30H c. AND AL,20H d. Or AL,20h	22. Consider that we pressed the number 6 button on the keyboard, then the value of AX is: MOV AB, 1 INT 21H ADD AL, 50H a) AX=8601 H b) AX=8701H c) AX=0601H d) AX=0186H
--	--

Consider the following data segment memory locations, answer questions from 23-25

0700:0102h	10	← X
0700:0103h	FF	
0700:0104h	4B	
0700:0105h	23	
0700:0106h	1F	← Y
0700:0107h	78	
0700:0108h	67	
0700:0109h	CD	

23. The value of BX after executing the following instruction is:
MOV BX, offset X
a) 700 b) 1F c) 0102 d) 0700:0102
24. The value of DI after executing the following instruction is:
MOV DI, [Y-3] c) 0700:0103 d) Error
a) 0103 b) 0FFh
25. The value of Y after executing the following instruction is:
LEA DI, Y
MOV [DI], 11
a) 08h b) 3 c) 11 d) 11h

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Question #3: (12 pts)

- A) Write An assembler codes to multiply each of memory location X by 4 using (using shift instructions) (4 pts.):
- B) Write an assembler codes to find number of locations that have values FFH ; assume you have 1 Kbyte memory locations to be tested. (4 POINTS)
- C) Write an assembler code to toggle bit#3 and bit #5 in each of memory locations X defined below X DB 65H,87,90,'a' bbb' (4 points)



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Question 2#:[14 points] Trace the following programs and determine the final value for each register:

1) What will be appears on the screen (3 pts)

```
STR DB 'AB', 0AH, 0AH, 'CD', '$', 'EF', 0DH, 0DH, 0AH, 'GH$'
```

```
MOV AH, 2  
MOV BH, 0  
MOV DX, 0102H  
INT 10H  
MOV AH, 9  
MOV DX, OFFSET STR  
INT 21H
```


2) X DB 0,1,7,8,7,5,7,8,6,9,5,9

```
MOV SI, 0  
MOV AL, 0  
MOV CX, 11  
L: ADD AL, X[SI]  
INC SI  
CMP X[SI], 8  
LOOPNE L  
PUSH CX  
L2: MOV X[SI], '*'  
INC SI  
LOOP L2  
POP CX
```

X[4]= AL= ;CX=

3)

```
MOV AX, 41ACH  
MOV BX, 8B19H  
MOV CL, 4  
XCHG AL, BH  
SHR AL, 1  
ADC BL, 1  
ROR BH, 1  
ROR AH, CL  
XOR CL, FFH  
AH= ; CL=
```

AL =

5. The value of RES after executing the following code is (2 PTS)

```
MOV DX, 0  
MOV CX, 3  
MOV DI, 0  
;assume CF=1  
L1: SBB DL, ARR[SI]
```

```
INC DI  
LOOP L1  
MOV RES, DL  
ARR DB 9, -6, 2  
RES DB ?
```

RES =

4)

```
ARRAY DB 1, 7, 7, 4, 6, 8, 5, 7, 0  
MOV AL, 0  
MOV CX, 4  
MOV BX, OFFSET ARRAY  
L: STC  
ADC AL, [BX]  
ADD BX, 2  
LOOP L  
MOV AH, [BX]  
AH= CF=
```

AL=

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9. The equivalent instructions of STOSW, assuming DF flag = 0. A. MOV [DI], AX; INC DI; INC DI B. MOV AX, [SI]; DEC SI C. MOV AX, [SI]; INC SI; INC SI D. MOV [DI], AX; DEC DI	10. The values of array B after executing the following code is: <pre> X DB 'ABCD' Y DB 5 DUP(10) CWD MOV SI, OFFSET X MOV DI, OFFSET Y MOV CX, 3 REP MOVSB MOV AL, 'F' STOSB </pre> A) Y=ABCDX B) Y=ABCDH C) Y=ABCDX D) Y=ABCF
11. Consider that we pressed the number 7 button on the keyboard, then the value of AX is: <pre> MOV AH, 1 INT 21H ADD AL, 50H </pre> A. AX=0701H B. AX=8701H C. AX=0187H D. AX=0107H	12. Which of the following code set the first and last bit of the register BX A. OR BX, 81H B. AND BX, 81H C. OR BX, 7EH D. AND BX, 7EH
13. Number of the times the instruction sequence below will loop before coming out of loop is <pre> MOV AL, FBH L1: INC AL JNZ L1 </pre> A) 5 B) 3 C) 256 D) 4	14. Find the content of AL and zero flag (ZF) after execution of the following code: <pre> MOV AL, 5BH TEST AL, 4 </pre> A. AL=0; ZF=1 B. AL=5BH; ZF=1 C. AL=5BH; ZF=0 D. AL=0; ZF=0
15. The result from the following declaration is: <pre> W DB 4 DUP(4,5) </pre> A) W=3,4,5 B) W[0]=W[2]=W[4] C) W=20 D) W=5	16. If the SP register contains the hexadecimal value 340H, what will be its value after executing the following program segment <pre> PUSH AX MOV BX, 25 PUSH BX MUL BX PUSH DX POP BX </pre> A) 344H B) 33CH C) 342H D) 338H

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9. The equivalent instructions of STOSW, assuming DF flag =0: A. MOV [DI],AX; INC DI, INC DI B. MOV AX,[SI]; DEC SI C. MOV AX,[SI]; INC SI, INC SI D. MOV [DI],AX; DEC DI	10. The values of array Y after executing the following code is: <pre>X DB 'ABCRR+CDK' Y DB 5 DUP('N') CLD MOV SI, OFFSET X MOV DI, OFFSET Y MOV CX, 3 REP MOVSB MOV AL, '#' STOSB</pre> A) Y=ABCR#N B) Y=ABC#N C) Y=ABC#RDK D) Y=ABC#*
11. Consider that we pressed the number 7 button on the keyboard, then the value of AX is: <pre>MOV AH, 1 INT 21H ADD AL, 50H</pre> A. AX=0701H B. AX=8701H C. AX=0187H D. AX=0107H	12. Which of the following code set the first and the last bit of the register BX A. OR BX, 81H B. AND BX, 81H C. OR BX, 7EH D. AND BX, 7EH
13. Number of the times the instruction sequence below will loop before coming out of loop is <pre>MOV AL, FBH INC AL JL L1</pre> A) 5 B) 3 C) 256 D) 4	14. Find the content of AL and zero flag (ZF) after execution of the following code: <pre>MOV AL, 58H TEST AL, 4</pre> A. AL=0; ZF=1 B. AL=58H; ZF=1 C. AL=58H; ZF=0 D. AL=0; ZF=0
15. The result from the following declaration is: <pre>W DB 4 DUP (4,5)</pre> A) 4,5 B) W[0]= W[2]= W[4] C) W=5 D) W=5	16. If the SP register contains the hexadecimal value 340H, what will be its value after executing the following program segment? <pre>PUSH AX MOV BX, 25 PUSH BX MUL BX PUSH DX POP BX</pre> A) 344H B) 33CH C) 342H D) 338H

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Question #1: select the correct answer (24 Pts)

1. LOPPNE instruction repeat a group of instruction while ----- A. (Cx=0) & (ZF=0) B. (CX=0) & (ZF≠0) C. (CX≠0) & (ZF=1) D) CX=0)&(ZF=0)	2. The result of division instruction DIV BL is stored in register pair A. AH:AL B. CX:AX C. AX:DX D. AX:CX
3. Find the content of memory location X after execution of the following code <pre>X DB ? MOV AX,18 MOV BL,3 DIV BL OR AH,30H MOV X,AH</pre> A. X='0' B. X=1 C. X='1' D. X=2	4. In the following instruction sequence, show the value of AL, in binary: <pre>MOV AL,FFH NEG AL XOR AL,FFH</pre> A. AL= 00000010B B. AL= 11111111B B. AL= 00000001B D. AL= 11111110B
5. Find the content of register AL after execution of the following code: <pre>MOV AL,0FH SHR AL,1 ADC AL,1</pre> A. 7 B. 8 C. 9 D. 6	6. The binary value of AL and carry flag (CF) after the following instructions have executed is ---- <pre>MOV CL,4 MOV AL,0FCH ROR AL,CL</pre> A. AL=11001111B, CF=0 B. AL=11001100B, CF=0 C. AL=11001111B, CF=1 D. AL=11001100B, CF=1
7. The value of CX and AL after executing the following code is: <pre>MOV AL,0 MOV CX,6 L1: ADD AX, CX LOOPNZ L1</pre> A. CX=5; AL=0 B. CX=0; AL=21 C. CX=5; AL=1 D. CX=3; AL=6	8. What is the value of the SP register after executing the following instruction, if the SP =FFFAH? <pre>POP AX DEC SP PUSH BX</pre> A) FFF7H B) FFF9H C) FFFBH D) FFF8H